

TECHNICAL ARCHITECTURE

Provable, not visible

How the attested market is built: what lives on chain, what does not, how intermediaries plug in without changing their systems, and how value is kept safe.

THE FIRST DECISION

The chain is a notary and a settlement rail, not a database

On chain	Why there	Off chain	Why not there
Commitments to records	Salted hashes, rolled into a root per asset per period	The records themselves	Rent rolls, leases, ledgers: confidential and erasable
Identity registry	Participants, agents, scopes, revocations	Mandates and credit boxes	Strategy; visible only through the enclave
Permission grants	Who may read what; every grant and revocation timestamped	The matching engine	Compute; inputs and outputs are committed, not the run
Attestation signatures	Which source signed which commitment, at what level	Models and extraction	Change weekly; only their outputs need lineage
Escrow and settlement state	Value moves atomically with title and tranche transfers	Documents	Large, sensitive, subject to retention rules
Audit log root	Every action's hash, so the log cannot be rewritten	Keys	Never anywhere but an HSM

The test for any new item: does a counterparty need to verify it later without trusting us? Then its commitment goes on chain. Does anyone need to read it? Then it stays off chain behind a grant. Nothing sensitive is ever permanent, which is also how right-to-erasure and confidentiality obligations are met.

COMMITMENTS AND DISCLOSURE

Three ways to prove a field without publishing it

1 · COMMIT

Salted hash per field

Each field is hashed with a random salt; fields roll up into a Merkle root per asset per period. The salt is essential: "NOI = 800,000" is guessable, so an unsalted hash is a leak.

2 · DISCLOSE

Grant the preimage

A grant hands a counterparty the value and salt for named fields. They verify against the root on chain. The grant itself is logged: who saw what, when, under whose permission.

3 · PROVE A PROPERTY

Range proof, no disclosure

A credit box needs "NOI $\geq$ X", not NOI. A zero-knowledge range proof answers that against the commitment. Used only where the double-blind market needs it; everything else is 1 and 2.

Documents follow the same pattern. A lease or a deed is hashed whole and its extracted fields hashed individually, so a counterparty can verify a field without the document, or the document without the fields.

Order of adoption. Commitments and grants are mature and cheap; build on them first. Range proofs are real but heavier: introduce them for credit boxes and mandate filters once the books are live.

TWO QUORUMS AND THE NETWORK

Consensus orders; attestation verifies. Keep them apart.

	Consensus quorum	Attestation quorum
Question	In what order did things happen, finally?	Is this fact true enough to act on?
Who votes	Validator nodes	Independent sources of the fact
Where it runs	The chain protocol	Application logic, off chain
What it writes	Blocks	Signatures over a commitment, and a level
Failure mode	Fork, halt	False fact with lineage
Fix	Protocol design	Source independence, cash, bonds

NETWORK TOPOLOGY

A permissioned network whose validators are the independent parties: notaries, auditors, depositaries, banks. Periodic roots anchored to a public chain, so finality rests on something no consortium member controls.

WHY THIS AND NOT PUBLIC

A bank's risk committee can approve validators it can name. Public anchoring gives the immutability; the permissioned set gives the accountability.

THE COMPONENTS

Eight services around one ledger

01 Identity

Participants, agents, credentials, scopes, revocation. Know Your Agent.

02 Ingestion

Connectors, extraction, canonical schema, lineage. Layer 01.

03 Record store

Encrypted per-tenant vaults, field-level keys, grants.

04 Attestation

Collects source signatures, computes quorum and level, commits.

05 Matching

Books, scoring, pair quotes, deferred acceptance. In an enclave.

06 Settlement

Escrow, DvP of title and cash, tranche transfer, notary binding.

07 Agent runtime

Policy engine, tool permissions, spend limits, human gates for acts.

08 Integration

Adapters, event bus, write-back. The one that decides adoption.

The ledger in the middle holds only what slide 02 lists. Every service writes commitments to it and reads grants from it; none of them stores state there.

Mapping to the layer model. 02 and 03 are Layer 01; 01, 04 and 06 are Layer 02; 05 and 07 are Layer 03; 08 cuts across all three.

ONE TRANSACTION, END TO END

What each step writes on chain, and what stays off

Step	Component	Off chain	On chain
List	Ingestion, attestation	Rent roll extracted; sources signed; level computed	Asset root, source signatures, level, seller grant policy
Diligence	Record store, agents	Buyer's agent reads granted fields; findings written	Grants and reads logged; DD findings root
Match	Matching enclave	Scoring, pair quotes, deferred acceptance run	Input roots, output root, enclave attestation
Finance	Matching, identity	Credit boxes evaluated; firm terms requested and returned	Term sheet commitment signed by the lender's agent
Settle	Settlement, identity	Closing set assembled; notary executes; bank releases	Escrow lock, DvP of title token and cash, tranche transfers
Operate	Ingestion, attestation	Rent, arrears, capex flow in monthly; bond reconciled	Period roots, level updates, bond release or forfeit

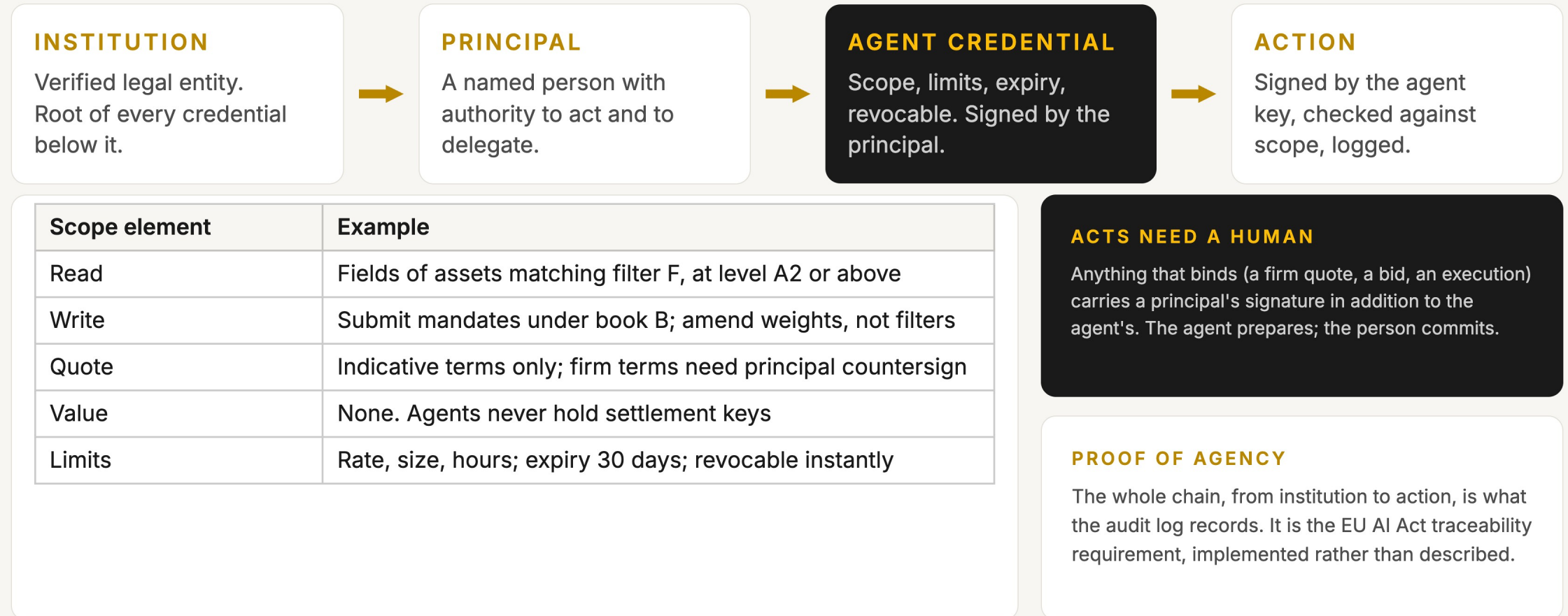
Nothing in the right column is readable. A regulator, an auditor or a counterparty can verify every step happened, in order, under permission, without seeing a rent, a mandate or a price.

The last row is where value compounds. Operate is not a step after the deal; it is the same pipeline running monthly, producing the realised state that the assurance engine prices against the claimed state.

Steps follow "What autonomy actually looks like", Infrastructure Deck slide 12.

IDENTITY AND DELEGATION

Every action traces to a person through a chain of scopes



THE MATCHING ENGINE

Neutrality as a property of the machine, not a promise

- 01 Deterministic.** Same books, same rules, same result. Any run can be replayed by an auditor from the committed inputs.
- 02 Committed in and out.** Roots of every mandate, asset and credit box go in; the root of the stable set comes out. Both on chain.
- 03 Runs in a trusted execution environment.** The books are decrypted only inside hardware that nobody, including us, can read from outside.
- 04 Code is attested.** The enclave proves which build it ran. Change the matching rule and the proof changes; participants can require a known build.

WHAT THIS PROVES

That the double-blind book was matched by known code on data no operator read. The neutrality line from the intermediary map stops being a governance claim and becomes a verifiable one.

WHAT IT DOES NOT

Enclaves have had hardware flaws. Treat it as one layer: keys still rotate, access is still logged, and a reveal to both parties is still gated by the hold window.

THE RECORD STORE

Data is encrypted per field; access is a grant, never a role

Element	Rule
Vaults	One per tenant; no shared tables; tenant keys never leave their vault's KMS
Field keys	Sensitive fields encrypted individually; a grant unlocks fields, not rows
Key custody	Hardware security modules; rotation on schedule and on any incident
Grants	Scoped, time-boxed, revocable; recorded on chain before the read happens
Erasure	Delete the field and its key; the on-chain commitment becomes unverifiable, which is the intent
Residency	Vault location follows the tenant's jurisdiction; commitments are location-free
Backups	Encrypted with separate keys; restore requires two custodians

WHY NOT ROLES

A role is standing access nobody remembers granting. A grant names the fields, the reason and the expiry, and is on the log before the first byte is read.

OPERATORS INCLUDED

Platform staff read under grants too. Support access is time-boxed and logged like any counterparty's; there is no admin key to the data.

INTEGRATION · THE PRINCIPLE

Valuable on day one with zero change to their workflow

STAGE 1 · SHADOW

Read only

Adapters read from what exists: ERP, property system, loan book, fund admin, and the PDFs and spreadsheets that were never a system. An attested copy appears beside their data. Nobody changes a habit.

STAGE 2 · WRITE BACK

Their format, our lineage

Attested outputs pushed into their system in the shape it expects: a reporting pack, a financing template, a covenant certificate. The value shows up where they already look.

STAGE 3 · WORKFLOW

They act on the platform

Their people or agents submit mandates, publish credit boxes, execute closings here because it is faster. Their system of record is still theirs; ours is the system of proof.

Why displacement fails. A core system is chosen for safety and record, not features, and replacing it is a multi-year programme with a career attached. Sitting beside it needs one sponsor and one connector.

Formats. Reuse the industry's exchange formats where they exist. The standard we set is the schema plus the attestation, never a new file type someone has to learn.

One canonical schema, many adapters, an event bus between

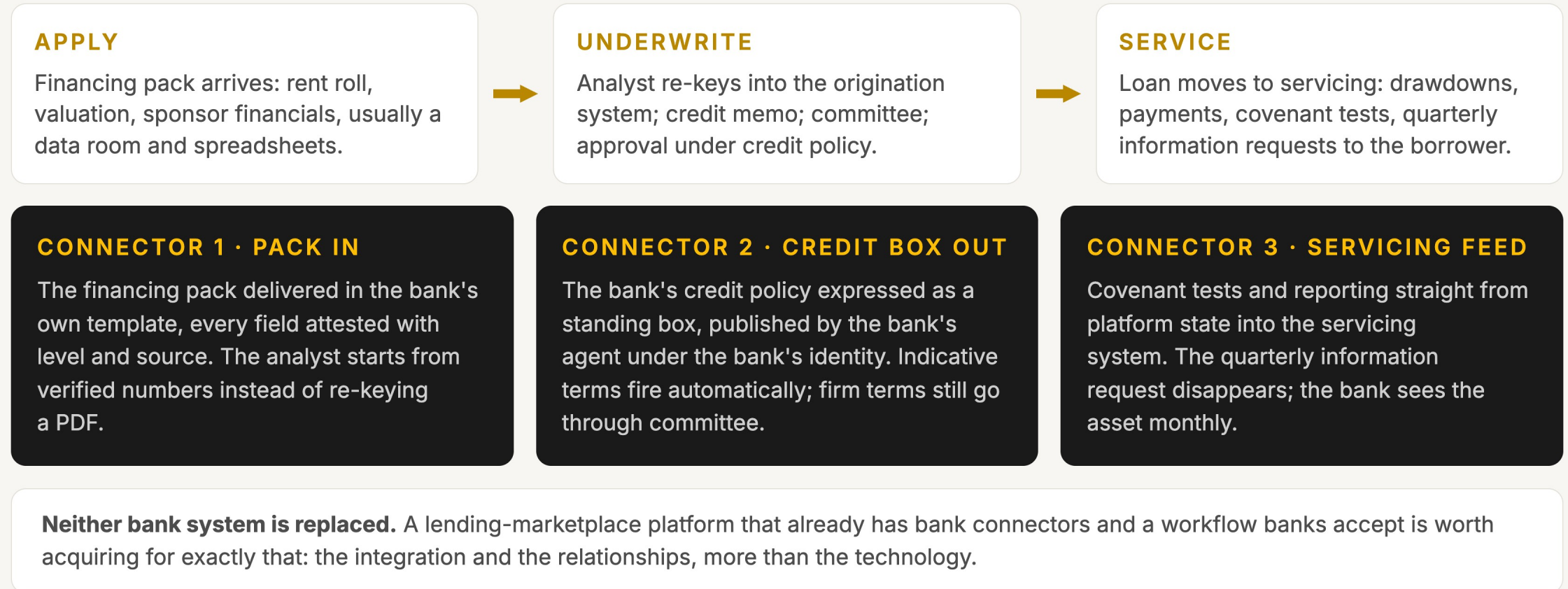
Source class	How it connects	What comes in	What goes back
ERP, accounting	API where offered, else scheduled export	Ledger, invoices, cash	Reconciled postings
Property management	API or export; often per country	Rent roll, leases, arrears	Attested rent roll
Loan origination, servicing	Template upload or API; bank-side connector	Credit policy, covenants	Financing packs, certificates
Fund administration	API or file exchange	NAV, register, cash	Reporting pack, proofs
Data rooms	Bulk pull, then extraction	Documents, DD reports	Structured findings
Registries, notaries	Portal or file where no API exists	Title state, deeds	Filing package, closing set
Files and email	Drop folder, mailbox, extraction pipeline	PDF, Excel, scans	Fields with source spans

Adapters are the product. The chain is generic; the adapter for a specific property system in a specific country is the thing a client cannot get elsewhere. Build them once, reuse them across every client on that system.

Every adapter is an untrusted input. It authenticates, rate-limits, validates against the schema, and its output is attested as one source, never as truth. A compromised adapter can lie; it cannot reach a quorum alone.

INTEGRATION · BANKS, WORKED

A loan today, and the three connectors that sit beside it



Generic description of commercial real estate lending flow; individual banks differ.

SECURITY · THE THREAT MODEL

Seven threats, ranked by what they have cost others

Threat	How it happens	Primary control
1 Key compromise	Phished, leaked or insider-held signing keys	No single key moves value; HSM custody; role separation
2 Contract defects	Bugs in escrow, token or settlement logic	Minimal on-chain logic; audits; formal checks; pause switch
3 Insider	Operator with standing access to data or keys	Grants not roles; two-person rule; full log; no admin key
4 Integration endpoint	Compromised adapter or partner system feeds lies	Adapter is one untrusted source; quorum; anomaly detection
5 Agent manipulation	Injected instructions in ingested documents	Content is data; scoped credentials; human gate on acts
6 Enclave or hardware	Side channel against the matching TEE	Defence in depth; key rotation; hold-window reveal
7 Availability	Validator loss, denial of service, outage	Validator diversity; public anchor; degraded read mode

Value and data are separated threats. A data breach is confidentiality and reputation; a key compromise is money gone. The architecture keeps them on different keys, different networks and different teams.

Rehearsed before there is anything to steal. Incident playbooks, key rotation drills and a pause exercise on the settlement contracts happen on testnet, then on the first small book.

Ranking reflects industry loss experience in general terms, not a measured study.

SECURITY · VALUE CUSTODY AND SETTLEMENT

Nine rules for money that moves on chain

1 No single key moves value. Multi-party custody with threshold signing; keys in HSMs; no hot key above a small operational float.

2 Role-separated keys. Deploy, upgrade, pause, settle and treasury are different keys held by different people.

3 Time-locks and velocity limits. Large settlements wait; daily outflow per counterparty is capped; anomalies pause automatically.

4 Allow-listed destinations only. Value settles only to addresses bound to a Know Your Agent identity. Unknown address, no transfer.

5 Circuit breakers. Every value contract can be paused by a quorum of holders; pausing is drilled, not theoretical.

6 Minimal, audited, slow-changing code. As little logic on chain as possible; external audit and formal checks; upgrades behind a governance delay.

7 Agents hold no value keys. Ever. They prepare settlement instructions; principals sign them.

8 Instructions from the record, never from a message. Payment details come from the attested closing set; an email cannot change them.

9 Recoverability designed in. Freeze and clawback on permissioned assets; custody insurance; monitoring that pages a human.

The stance: assume a key will be compromised and design so that one compromised key can do nothing on its own, can do only a little with a second, and can be reversed with a third. Most large on-chain losses would have been stopped by rules 1, 3 and 4 alone.

SECURITY · THE SETTLEMENT ASSET

Use someone else's money first; issue your own last

Option	What it is	We carry	When
Tokenised deposits	A bank's own liability on chain, for its clients	Integration only	Now, with the escrow bank
Regulated stablecoin	Licensed issuer, reserves, redemption at par	Counterparty selection	Now, for cross-bank settlement
Own issuance	We become the licensed issuer with reserves	Licence, reserves, redemption, audit	Only with volume that justifies a bank-grade function

PERMISSIONED PROPERTIES ARE FEATURES

Allow-list, freeze and clawback are what make a theft recoverable. The washed-out losses on open rails happen where those controls do not exist. In an institutional settlement asset they are requirements, not compromises.

WHAT ISSUING ACTUALLY MEANS

A regulated issuer holds reserves at a bank, redeems at par on demand, is audited on both, and answers to the regulator for every freeze. It is a business, and a different one. Earn the volume first.

Float, revisited. Escrow balances and settlement float sit with the escrow bank under its licence, earning yield that can be shared. That is most of the economic benefit of a native coin with none of the issuer obligations.

Regulatory regimes for e-money and stablecoins differ by jurisdiction; specifics to be confirmed with counsel.

THE BUILD SEQUENCE

Commit first. Connect second. Match third. Settle last.

NOW

- Salted commitments, roots, grant log
- Identity and agent credentials
- Record store with field keys and HSM
- Adapters for the internal portfolio

Decide: the first three named validators outside the group.

NEXT

- Permissioned network with named validators
- Bank connectors: pack in, box out
- Matching engine in an attested enclave
- Escrow contracts on tokenised deposits

Decide: the escrow bank and its tokenised-deposit capability.

LATER

- Range proofs for double-blind filters
- Atomic DvP with notary execution
- Public anchoring, validator expansion
- Own settlement asset, if ever

Decide: build or acquire the bank-side connectors.

Discussion

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